ENGINEERING ECONOMICS (CE 655)

Chapter 4: Comparative Analysis of Alternatives

Questions & Answers — TU Past Exam Papers (2070–2082)

CHAPTER 4 SYLLABUS OVERVIEW (6 Hours)

(a) Introduction — Mutually Exclusive vs. Independent Alternatives, Do-Nothing Option

(b) Comparing Equal-Life Alternatives using PW, AW, FW methods

(c) Comparing Unequal-Life Alternatives — Repeatability (LCM) Assumption, Co-terminated (Study Period) Assumption

(d) Incremental Analysis — Incremental IRR (ΔIRR), Incremental B/C Ratio

(e) Capitalized Cost Comparison

(f) Cost-Only Alternatives

(g) Selection among multiple mutually exclusive alternatives

SECTION A: Fundamental Concepts — Comparative Analysis

Q1. What do you mean by comparative analysis? Explain mutually exclusive and independent alternatives with examples. [2081 Ashwin Q4a, 2075 Bhadra Q4b — 4 Marks]

Answer:

Comparative Analysis

Comparative analysis is the process of systematically evaluating two or more engineering project alternatives to identify the one that is most economically desirable. The goal is to select the best alternative based on a defined decision criterion such as maximising NPW, minimising cost, or achieving the highest IRR, subject to the constraint that the chosen project meets or exceeds the MARR.

All alternatives must be compared using the same evaluation period, the same interest rate (MARR), and the same viewpoint to ensure a valid comparison.

Mutually Exclusive Alternatives

A set of alternatives is mutually exclusive when selecting one alternative automatically prevents the selection of any other. Only one option from the set can be implemented.

Characteristics:

- Selection of one precludes selection of all others.
- Each alternative solves the same problem or meets the same need.
- Comparison is made using incremental analysis or by comparing equivalent worths.

Decision criterion: Select the alternative with the highest NPW (or AW or FW) that is ≥ 0, or the alternative whose incremental investment over the next-best option earns at least MARR.

Example: A municipality must choose ONE of three bridge designs — pre-stressed concrete, steel truss, or suspension bridge — to cross a river. Selecting the concrete bridge means the steel truss and suspension options are rejected. These are mutually exclusive alternatives.

Independent Alternatives

A set of alternatives is independent when selecting one does not affect the selection of others. Any subset (including none or all) can be selected, subject to a budget constraint.

Characteristics:

- Each alternative is evaluated on its own merits (accept if NPW ≥ 0).
- Alternatives solve different problems or serve different purposes.
- Multiple alternatives can be selected simultaneously.

Example: A company has four independent projects: (1) purchase a new computer server, (2) upgrade factory lighting, (3) install solar panels, (4) buy a delivery van. Each project independently earns a return. If the budget permits, all four can be selected. Rejecting the solar panel project does not prevent choosing the delivery van.

Basis	Mutually Exclusive	Independent
Selection	Only ONE can be chosen	Any number can be chosen
Evaluation method	Incremental analysis / Max NPW	Accept each if NPW ≥ 0
Relationship	Alternatives solve same problem	Alternatives solve different problems
Budget constraint	Always implied	Often applies
Example	Three designs for one bridge	Four unrelated equipment upgrades

Do-Nothing Alternative

In any comparative analysis, the 'Do Nothing' option must always be considered. It represents maintaining the status quo — making no investment. The Do-Nothing alternative has zero NPW and zero cost. Any selected alternative must perform better than doing nothing (NPW ≥ 0) to be justified.

Q2. What are the two fundamental assumptions used when comparing alternatives with unequal lives? Explain each with example. [2080 Chaitra Q4a, 2078 Chaitra Q4a — 4 Marks]

Answer:

When comparing two or more mutually exclusive alternatives with different useful lives (e.g., n=5 years vs. n=8 years), a common study period must be established. Two assumptions are used:

Assumption 1: Repeatability Assumption (LCM Method)

The repeatability assumption states that each alternative will be repeated (renewed) in identical fashion at the end of its life until all alternatives have reached the same total time horizon — the Least Common Multiple (LCM) of their individual lives.

Conditions for validity:

- The need for the project continues over the LCM period.
- Each alternative can be repeated with the same cash flows (adjusted for inflation if necessary).
- The LCM period is not unreasonably long (e.g., LCM of 3 and 4 = 12 years is acceptable; LCM of 7 and 11 = 77 years is impractical).

Implementation: Cash flows are replicated over the LCM period and compared using PW at the common end point. Alternatively, since all alternatives are repeated, AW over one life cycle is a valid comparison measure — the AW of one cycle is the same as the AW over the entire LCM period.

Example: Alternative A has a 4-year life; Alternative B has a 6-year life. LCM = 12 years. Alternative A is repeated 3 times; Alternative B is repeated 2 times. PW of A over 12 years = PW of three successive cycles of A. Or simply compare AW(A) over 4 years with AW(B) over 6 years.

Assumption 2: Co-terminated Assumption (Study Period / Planning Horizon Method)

The co-terminated assumption truncates all alternatives to the same analysis period — the study period or planning horizon — chosen by the analyst. Alternatives with lives shorter than the study period are assumed to be repeated (or the service gap is filled by other means). Alternatives with lives longer than the study period are terminated early, and a market value (salvage value) is estimated at the end of the study period.

Conditions for validity:

- A specific planning period is mandated (e.g., contract period, project phase duration).
- The LCM is impractically long.
- Technology is changing rapidly, making long-term repetition unrealistic.

Example: Alternative A has a 3-year life; Alternative B has a 9-year life. Study period = 5 years. Alternative A is implemented for years 1-3, then a shortened 2-year version fills years 4-5. Alternative B is used for 5 years and sold at its estimated market value at year 5.

Basis	Repeatability (LCM) Assumption	Co-terminated (Study Period) Assumption
Study period	LCM of all project lives	Fixed by analyst or management
Shorter-life projects	Repeated until LCM	Repeated to fill study period
Longer-life projects	Naturally repeats for fewer cycles	Terminated early; market value estimated
Best comparison metric	AW over one life cycle	PW over the study period
When appropriate	Lives are not too dissimilar; need continues	Fixed horizon; impractical LCM; rapid tech change

SECTION B: Comparing Equal-Life Alternatives

Q3. Compare the following two mutually exclusive alternatives using the PW method. MARR = 10% per year. [2082 Kartik Q4a, 2081 Chaitra Q4b — 6 Marks]

Answer:

	Alternative A	Alternative B
Initial Cost (Rs.)	30,000	50,000
Annual Revenue (Rs.)	9,000	13,500
Annual Expense (Rs.)	2,000	3,000
Salvage Value (Rs.)	5,000	8,000
Useful Life (years)	10	10

MARR = 10%, n = 10 years (equal lives — direct comparison using PW)

PW of Alternative A:

$$PW_A = -30{,}000 + (9{,}000-2{,}000)(P/A,10\%,10) + 5{,}000(P/F,10\%,10)$$

$$(P/A,10\%,10) = 6.1446, \quad (P/F,10\%,10) = 0.3855$$

$$PW_A = -30{,}000 + 7{,}000(6.1446) + 5{,}000(0.3855)$$
$$= -30{,}000 + 43{,}012 + 1{,}928 = Rs.\ 14{,}940$$

PW of Alternative B:

$$PW_B = -50{,}000 + (13{,}500-3{,}000)(P/A,10\%,10) + 8{,}000(P/F,10\%,10)$$
$$PW_B = -50{,}000 + 10{,}500(6.1446) + 8{,}000(0.3855)$$
$$= -50{,}000 + 64{,}518 + 3{,}084 = Rs.\ 17{,}602$$

Both PW_A and PW_B > 0. Both alternatives exceed MARR. Since PW_B (Rs. 17,602) > PW_A (Rs. 14,940), select Alternative B.

Q4. Use AW method to select the best alternative at MARR = 12% per year. [2079 Jestha Q4a, 2077 Chaitra Q4a — 6 Marks]

Answer:

	Machine X	Machine Y	Machine Z
Capital Cost (Rs.)	50,000	80,000	1,10,000
Annual O&M (Rs.)	12,000	8,000	6,000
Annual Revenue (Rs.)	28,000	32,000	38,000
Salvage Value (Rs.)	8,000	12,000	20,000
Life (years)	5	5	5

MARR = 12%, n = 5 years (equal lives)

$$(A/P,12\%,5) = 0.27741, \quad (A/F,12\%,5) = 0.15741$$

AW of Machine X:

$$AW_X = -50{,}000(A/P,12\%,5) + 8{,}000(A/F,12\%,5) + (28{,}000-12{,}000)$$
$$= -50{,}000(0.27741) + 8{,}000(0.15741) + 16{,}000$$
$$= -13{,}871 + 1{,}259 + 16{,}000 = Rs.\ 3{,}388$$

AW of Machine Y:

$$AW_Y = -80{,}000(0.27741) + 12{,}000(0.15741) + (32{,}000-8{,}000)$$
$$= -22{,}193 + 1{,}889 + 24{,}000 = Rs.\ 3{,}696$$

AW of Machine Z:

$$AW_Z = -1{,}10{,}000(0.27741) + 20{,}000(0.15741) + (38{,}000-6{,}000)$$
$$= -30{,}515 + 3{,}148 + 32{,}000 = Rs.\ 4{,}633$$

All AW values are positive (all meet MARR). AW_Z (Rs. 4,633) is highest. Select Machine Z.

Q5. A company needs to purchase one of the following pumps. Using AW method at MARR = 10%, which pump should be selected? [2079 Ashwin Q4a — 5 Marks]

Answer:

	Pump A	Pump B
First Cost (Rs.)	70,000	95,000
Annual Operating Cost (Rs.)	8,000	5,000
Salvage Value (Rs.)	10,000	15,000
Life (years)	8	8

Cost-only comparison (no revenue given). Select the alternative with the least negative AW (lowest cost).

$$(A/P,10\%,8) = 0.18744, \quad (A/F,10\%,8) = 0.08744$$

AW of Pump A (cost):

$$AW_A = -70{,}000(A/P,10\%,8) + 10{,}000(A/F,10\%,8) - 8{,}000$$
$$= -70{,}000(0.18744) + 10{,}000(0.08744) - 8{,}000$$
$$= -13{,}121 + 874 - 8{,}000 = -Rs.\ 20{,}247$$

AW of Pump B (cost):

$$AW_B = -95{,}000(0.18744) + 15{,}000(0.08744) - 5{,}000$$
$$= -17{,}807 + 1{,}311 - 5{,}000 = -Rs.\ 21{,}496$$

AW_A (-Rs. 20,247) > AW_B (-Rs. 21,496) in algebraic terms (less negative = lower annual cost). Select Pump A.

Q6. Select the best alternative using PW method at MARR = 12%. All alternatives have a life of 6 years. [2078 Baishakh Q4b — 6 Marks]

Answer:

	Alt. 1	Alt. 2	Alt. 3
Investment (Rs.)	40,000	75,000	1,00,000
Annual Benefit (Rs.)	18,000	28,000	38,000
Annual Cost (Rs.)	5,000	8,000	12,000
Salvage Value (Rs.)	3,000	5,000	8,000
Life (years)	6	6	6

MARR = 12%, n = 6 years

$$(P/A,12\%,6) = 4.1114, \quad (P/F,12\%,6) = 0.5066$$

PW Calculations:

$$PW_1 = -40{,}000 + (18{,}000-5{,}000)(4.1114) + 3{,}000(0.5066)$$
$$= -40{,}000 + 13{,}000(4.1114) + 1{,}520 = -40{,}000 + 53{,}448 + 1{,}520 = Rs.\ 14{,}968$$

$$PW_2 = -75{,}000 + (28{,}000-8{,}000)(4.1114) + 5{,}000(0.5066)$$
$$= -75{,}000 + 20{,}000(4.1114) + 2{,}533 = -75{,}000 + 82{,}228 + 2{,}533 = Rs.\ 9{,}761$$

$$PW_3 = -1{,}00{,}000 + (38{,}000-12{,}000)(4.1114) + 8{,}000(0.5066)$$

$$= -1,00,000 + 26,000(4.1114) + 4,053 = -1,00,000 + 1,06,896 + 4,053 = \text{Rs. } 10,949$$

PW_1 (Rs. 14,968) is the highest. Select Alternative 1.

Note: Although Alt. 1 has the smallest investment, it generates the best return relative to cost at MARR=12%. This illustrates why NPW — not just return percentage or revenue — must be the decision criterion.

SECTION C: Comparing Unequal-Life Alternatives

Q7. Compare the following alternatives using AW method (repeatability assumption). MARR = 10% per year. [2082 Kartik Q4b, 2080 Ashwin Q4a — 6 Marks]

Answer:

	Alternative X	Alternative Y
First Cost (Rs.)	40,000	65,000
Annual Net Benefit (Rs.)	12,000	15,000
Salvage Value (Rs.)	5,000	10,000
Useful Life (years)	4	6

MARR = 10%. Unequal lives (4 vs 6 years). LCM = 12 years.

Since we use the AW method with repeatability, we compute AW over ONE life cycle for each alternative — this AW is valid for the entire LCM period.

$$(A/P,10\%,4)=0.31547, \quad (A/F,10\%,4)=0.21547$$

$$(A/P,10\%,6)=0.22961, \quad (A/F,10\%,6)=0.12961$$

AW of Alternative X (over 4 years):

$$AW_X = -40,000(A/P,10\%,4) + 5,000(A/F,10\%,4) + 12,000$$

$$= -40,000(0.31547) + 5,000(0.21547) + 12,000$$

$$= -12,619 + 1,077 + 12,000 = \text{Rs. } 458$$

AW of Alternative Y (over 6 years):

$$AW_Y = -65,000(A/P,10\%,6) + 10,000(A/F,10\%,6) + 15,000$$

$$= -65,000(0.22961) + 10,000(0.12961) + 15,000$$

$$= -14,925 + 1,296 + 15,000 = \text{Rs. } 1,371$$

AW_Y (Rs. 1,371) > AW_X (Rs. 458). Both exceed MARR (AW > 0). Select Alternative Y.

Verification using LCM (PW method): X repeats 3 times; Y repeats 2 times over 12 years.

$$PW_X(12yr) = AW_X \times (P/A,10\%,12) = 458 \times 6.8137 = \text{Rs. } 3,120$$

$$PW_Y(12yr) = AW_Y \times (P/A,10\%,12) = 1,371 \times 6.8137 = \text{Rs. } 9,342$$

PW comparison confirms: Select Y.

Q8. Using PW method and repeatability assumption, compare these alternatives at MARR = 8%. [2079 Chaitra Q4a — 5 Marks]

Answer:

	Alt. A	Alt. B
First Cost (Rs.)	25,000	45,000
Annual O&M Cost (Rs.)	4,000	2,500
Salvage Value (Rs.)	3,000	6,000
Life (years)	3	6

MARR = 8%. LCM(3, 6) = 6 years. Alt. A is repeated twice; Alt. B runs once.

```
(P/F,8%,3) = 0.7938,   (P/F,8%,6) = 0.6302
(P/A,8%,3) = 2.5771,   (P/A,8%,6) = 4.6229
```

PW of Alt. A over 6 years (A repeated at t=0 and t=3):

Cycle 1: t=0 to t=3

```
PW_A1 = -25,000 - 4,000(P/A,8%,3) + 3,000(P/F,8%,3)
      = -25,000 - 10,308 + 2,381 = -32,927
```

Cycle 2 starts at t=3, so PW at t=0:

```
PW_A2 = (-25,000 - 10,308 + 2,381) × (P/F,8%,3) = -32,927 × 0.7938 = -26,140
PW_A_total = -32,927 + (-26,140) = -Rs. 59,067
```

PW of Alt. B over 6 years (runs once):

```
PW_B = -45,000 - 2,500(P/A,8%,6) + 6,000(P/F,8%,6)
     = -45,000 - 11,557 + 3,781 = -Rs. 52,776
```

Cost comparison: PW_B (-Rs. 52,776) < PW_A (-Rs. 59,067) in absolute cost. Alt. B costs less over the LCM period. Select Alternative B.

(Lower cost = less negative PW = better for cost-only alternatives)

Q9. Compare the following cost alternatives using the AW method at MARR = 10%. [2081 Chaitra Q4a — 6 Marks]

Answer:

	Machine P	Machine Q	Machine R
Initial Cost (Rs.)	60,000	1,00,000	75,000
Annual Maintenance (Rs.)	8,000	4,000	6,000
Salvage Value (Rs.)	10,000	15,000	12,000
Life (years)	3	6	4

MARR = 10%. Unequal lives. Use AW method (repeatability assumption — AW over each cycle is independent of LCM).

```
(A/P,10%,3)=0.40211,   (A/F,10%,3)=0.30211
(A/P,10%,6)=0.22961,   (A/F,10%,6)=0.12961
(A/P,10%,4)=0.31547,   (A/F,10%,4)=0.21547
```

AW of Machine P (3 years):

```
AW_P = -60,000(0.40211) + 10,000(0.30211) - 8,000
     = -24,127 + 3,021 - 8,000 = -Rs. 29,106
```

AW of Machine Q (6 years):

```
AW_Q = -1,00,000(0.22961) + 15,000(0.12961) - 4,000
     = -22,961 + 1,944 - 4,000 = -Rs. 25,017
```

AW of Machine R (4 years):

```
AW_R = -75,000(0.31547) + 12,000(0.21547) - 6,000
     = -23,660 + 2,586 - 6,000 = -Rs. 27,074
```

AW_Q (-Rs. 25,017) is the least negative (lowest annual cost). Select Machine Q.

Q10. A project with 15-year study period must choose between two alternatives. Compare using PW with co-terminated assumption at MARR = 10%. [2080 Chaitra Q4a — 6 Marks]

Answer:

	Alternative A	Alternative B
First Cost (Rs.)	1,20,000	1,80,000
Annual Net Revenue (Rs.)	35,000	45,000
Salvage Value (Rs.)	20,000	30,000
Useful Life (years)	5	15

Study period = 15 years, MARR = 10%.

Alt. A (5-year life): Repeated 3 times over 15 years (at t=0, t=5, t=10). Salvage of Rs. 20,000 received at t=5 and t=10 only for intermediate cycles; final cycle ends naturally at t=15.

Alt. B (15-year life): Runs once, ends exactly at study period.

PW of Alternative A (three 5-year cycles):

```
(P/A,10%,5)=3.7908,  (P/F,10%,5)=0.6209,  (P/F,10%,10)=0.3855,
(P/F,10%,15)=0.2394
```

Cycle 1 (t=0–5): -1,20,000 + 35,000(P/A,10%,5) + 20,000(P/F,10%,5)

```
= -1,20,000 + 35,000(3.7908) + 20,000(0.6209) = -1,20,000 + 1,32,678 +
12,418 = +25,096
```

Cycle 2 (t=5–10) — PW at t=5, then discount to t=0:

```
PW@t5 = same = +25,096
PW_cycle2@t=0 = 25,096 × 0.6209 = 15,587
```

Cycle 3 (t=10–15) — PW at t=10, then discount to t=0:

```
PW_cycle3@t=0 = 25,096 × 0.3855 = 9,675
PW_A_total = 25,096 + 15,587 + 9,675 = Rs. 50,358
```

PW of Alternative B (single 15-year cycle):

```
(P/A,10%,15)=7.6061
```

```
PW_B = -1,80,000 + 45,000(7.6061) + 30,000(0.2394)
     = -1,80,000 + 3,42,275 + 7,182 = Rs. 1,69,457
```

PW_B (Rs. 1,69,457) >> PW_A (Rs. 50,358). Select Alternative B.

Q11. A municipality must choose between two water supply systems. Compare using AW method at MARR = 8%. Identify the LCM study period. [2077 Chaitra Q4b — 6 Marks]

Answer:

	System Alpha	System Beta
Capital Cost (Rs. Lakhs)	80	120
Annual O&M (Rs. Lakhs)	5	3
Salvage Value (Rs. Lakhs)	10	15
Useful Life (years)	6	9

MARR = 8%. LCM(6, 9) = 18 years. Using AW (repeatability), compute AW for one cycle each.

```
(A/P,8%,6)=0.21632,   (A/F,8%,6)=0.13632
(A/P,8%,9)=0.16008,   (A/F,8%,9)=0.08008
```

AW of System Alpha (one 6-year cycle):

```
AW_Alpha = -80(0.21632) + 10(0.13632) - 5 = -17.306 + 1.363 - 5 = -Rs.
                                20.943 Lakhs/yr
```

AW of System Beta (one 9-year cycle):

```
AW_Beta  = -120(0.16008) + 15(0.08008) - 3 = -19.210 + 1.201 - 3 = -Rs.
                                21.009 Lakhs/yr
```

AW_Alpha (-Rs. 20.943 L) is slightly less costly than AW_Beta (-Rs. 21.009 L). Select System Alpha.

The cost difference is marginal (Rs. 0.066 Lakhs/year ≈ Rs. 6,600/year). In practice, non-economic factors such as reliability, maintenance capacity, and future demand should also be considered.

SECTION D: Incremental Analysis — Incremental IRR (ΔIRR)

Q12. Explain the procedure for incremental IRR analysis for comparing mutually exclusive alternatives. Why is incremental analysis necessary? [2081 Ashwin Q4b, 2074 Bhadra Q4a — 4 Marks]

Answer:

Why Incremental IRR Analysis is Necessary

When comparing mutually exclusive alternatives using IRR, it is incorrect to simply select the alternative with the highest IRR. A high-IRR alternative may require a smaller initial investment and create less total value than a lower-IRR alternative with a larger investment. This is the inconsistent ranking problem discussed in Chapter 3.

Example: Alt. A costs Rs. 10,000 with IRR = 35%. Alt. B costs Rs. 50,000 with IRR = 22%. At MARR = 10%, NPW(A) = Rs. 2,000 while NPW(B) = Rs. 18,000. IRR ranks A first, but B creates far more

value. Incremental analysis on (B-A) determines whether the extra Rs. 40,000 invested in B over A earns at least MARR.

Procedure for Incremental IRR Analysis:

1. Order Alternatives: Arrange all alternatives in ascending order of initial investment (including the Do-Nothing option at Rs. 0 investment).
2. Verify Each Alternative: Compute the IRR of each alternative individually. Eliminate any alternative whose own IRR < MARR (it cannot even earn the minimum required return on its own).
3. Select the Defender: Start with the Do-Nothing option (or the lowest-cost alternative that meets MARR) as the current best ('Defender').
4. Compute Incremental Cash Flows: Subtract the Defender's cash flows from the next alternative ('Challenger') in the ordered list to get the incremental cash flow stream (Δ CF = CF_Challenger - CF_Defender).
5. Find Incremental IRR (ΔIRR): Solve for the interest rate that makes the NPW of the incremental cash flow = 0. This is the ΔIRR.
6. Compare ΔIRR to MARR:
   - If ΔIRR ≥ MARR: The additional investment in the Challenger is justified. The Challenger becomes the new Defender.
   - If ΔIRR < MARR: The additional investment is NOT justified. The Defender remains the current best.
7. Repeat: Continue the pairwise comparison with the next Challenger until all alternatives are evaluated.
8. Final Decision: The last surviving Defender is the best alternative.

Key Rule: The selection is always between two alternatives at a time. The extra investment must justify itself by earning at least MARR.

Q13. Using incremental IRR analysis, determine which alternative should be selected. MARR = 12%. [2082 Kartik Q5a, 2079 Jestha Q5b — 8 Marks]

Answer:

	Alt. A (Do Nothing)	Alt. B	Alt. C	Alt. D
Initial Cost (Rs.)	0	20,000	30,000	45,000
Annual Net Benefit (Rs.)	0	5,000	8,000	11,000
Salvage Value (Rs.)	0	2,000	3,500	5,000
Life (years)	—	8	8	8

MARR = 12%, n = 8 years for all alternatives.

(P/A,12%,8)=4.9676, (P/F,12%,8)=0.4039

Step 1: Compute individual IRR of each alternative

Alt. B: PW = -20,000 + 5,000(P/A,i,8) + 2,000(P/F,i,8) = 0

Try i=18%: PW = -20,000+5,000(3.8030)+2,000(0.2660) = -20,000+19,015+532 = -453

Try i=17%: PW = -20,000+5,000(3.9224)+2,000(0.2848) = -20,000+19,612+570 = +182

IRR_B = 17+1×182/(182+453) = 17+0.29 = 17.3% > MARR(12%). B is viable.

Alt. C: PW = -30,000 + 8,000(P/A,i,8) + 3,500(P/F,i,8) = 0

Try i=19%: PW = -30,000+8,000(3.7057)+3,500(0.2487) = -30,000+29,646+871 = +517

Try i=20%: PW = -30,000+8,000(3.6046)+3,500(0.2326) = -30,000+28,837+814 = -349

IRR_C = 19+1×517/(517+349) = 19+0.60 = 19.6% > MARR. C is viable.

Alt. D: PW = -45,000 + 11,000(P/A,i,8) + 5,000(P/F,i,8) = 0

Try i=20%: PW = -45,000+11,000(3.6046)+5,000(0.2326) = -45,000+39,651+1,163 = -4,186

Try i=15%: PW = -45,000+11,000(4.4873)+5,000(0.3269) = -45,000+49,360+1,635 = +5,995

Try i=18%: PW = -45,000+11,000(3.8030)+5,000(0.2660) = -45,000+41,833+1,330 = -1,837

IRR_D = 15+3×5,995/(5,995+4,186) = 15+1.76 ≈ 16.76%... try narrower: i=17%:

PW@17% = -45,000+11,000(3.9224)+5,000(0.2848) = -45,000+43,146+1,424 = -430

IRR_D = 15+2×5,995/(5,995+430) ~ closer to 16.9%

All three alternatives exceed MARR (12%). Order: B < C < D by initial investment.

Step 2: Incremental Analysis — (B vs. Do Nothing)

Defender = Do Nothing. Challenger = B. Incremental CF = CF_B - CF_DN = CF_B.

ΔIRR(B vs DN) = IRR_B = 17.3% > MARR (12%). Accept B; B becomes new Defender.

Step 3: Incremental Analysis — (C vs. B)

ΔCF at t=0: -30,000 - (-20,000) = -10,000

ΔCF at t=1-8: 8,000 - 5,000 = 3,000/year

ΔCF at t=8 (salvage): 3,500 - 2,000 = 1,500

Find ΔIRR(C-B):

PW_Δ = -10,000 + 3,000(P/A,i,8) + 1,500(P/F,i,8) = 0

Try i=25%: PW = -10,000+3,000(3.3289)+1,500(0.1678) = -10,000+9,987+252 = +239

Try i=26%: PW = -10,000+3,000(3.2407)+1,500(0.1594) = -10,000+9,722+239 = -39

ΔIRR(C-B) = 25+1×239/(239+39) = 25+0.86 = 25.86%

ΔIRR(C-B) = 25.86% > MARR (12%). The extra Rs. 10,000 in C over B earns 25.86%. Accept C; C becomes new Defender.

Step 4: Incremental Analysis — (D vs. C)

ΔCF at t=0: -45,000 - (-30,000) = -15,000

ΔCF at t=1-8: 11,000 - 8,000 = 3,000/year

ΔCF at t=8 (salvage): 5,000 - 3,500 = 1,500

PW_Δ = -15,000 + 3,000(P/A,i,8) + 1,500(P/F,i,8) = 0

Try i=12%: PW = -15,000+3,000(4.9676)+1,500(0.4039) = -15,000+14,903+606 = +509

Try i=13%: PW = -15,000+3,000(4.7988)+1,500(0.3762) = -15,000+14,396+564 = -40

ΔIRR(D-C) = 12+1×509/(509+40) = 12+0.93 = 12.93%

ΔIRR(D-C) = 12.93% > MARR (12%). The extra Rs. 15,000 in D over C earns 12.93%. Accept D; D becomes new Defender.

No more challengers. Final Answer: Select Alternative D (the largest investment).

This confirms: using incremental IRR, the best alternative is not necessarily the one with the highest individual IRR (B has the highest IRR at ~17.3%), but the one that maximises total economic value (D).

Q14. Use incremental IRR to find the best alternative from the following. MARR = 15%. [2080 Ashwin Q4b — 6 Marks]

Answer:

EOY	Alt. 1	Alt. 2	Alt. 3
0	-20,000	-35,000	-55,000
1	8,000	15,000	22,000
2	8,000	15,000	22,000
3	8,000	15,000	22,000
4	8,000	15,000	22,000
5	8,000+2,000	15,000+3,000	22,000+5,000

MARR = 15%. All alternatives have 5-year life. Annual benefit + salvage value at year 5 shown.

(P/A,15%,5)=3.3522, (P/F,15%,5)=0.4972

Step 1: Individual IRRs

Alt 1: PW = -20,000+8,000(P/A,i,5)+2,000(P/F,i,5) = 0

Try i=28%: PW = -20,000+8,000(2.7594)+2,000(0.3916) = -20,000+22,075+783 = +2,858

Try i=35%: PW = -20,000+8,000(2.3852)+2,000(0.2230) = -20,000+19,082+446 = -472

IRR_1 = 28+7×2,858/(2,858+472) = 28+6.0 = 34.0%

Alt 2: PW = -35,000+15,000(P/A,i,5)+3,000(P/F,i,5) = 0

Try i=38%: PW = -35,000+15,000(2.2200)+3,000(0.1985) = -35,000+33,300+596 = -1,104

Try i=35%: PW = -35,000+15,000(2.3852)+3,000(0.2230) = -35,000+35,778+669 = +1,447

IRR_2 = 35+3×1,447/(1,447+1,104) = 35+1.7 = 36.7%

Alt 3: PW = -55,000+22,000(P/A,i,5)+5,000(P/F,i,5) = 0

Try i=30%: PW = -55,000+22,000(2.6427)+5,000(0.2693) = -55,000+58,139+1,347 = +4,486

Try i=38%: PW = -55,000+22,000(2.2200)+5,000(0.1985) = -55,000+48,840+993 = -5,167

IRR_3 = 30+8×4,486/(4,486+5,167) = 30+3.7 = 33.7%

All three IRRs > MARR (15%). All viable. Order by investment: 1 < 2 < 3.

Step 2: ΔIRR (Alt. 2 vs Alt. 1)

ΔCF(t=0): -35,000-(-20,000) = -15,000

ΔCF(t=1-4): 15,000-8,000 = 7,000/year

ΔCF(t=5): (15,000+3,000)-(8,000+2,000) = 8,000

PW_Δ = -15,000+7,000(P/A,i,4)+8,000(P/F,i,5) = 0

Try i=40%: PW=-15,000+7,000(2.0352)+8,000(0.1859)=-15,000+14,246+1,487=+733

Try i=45%: PW=-15,000+7,000(1.9209)+8,000(0.1560)=-15,000+13,446+1,248=-306

ΔIRR(2-1) = 40+5×733/(733+306) = 40+3.5 = 43.5% > MARR (15%). Accept Alt. 2.

Step 3: ΔIRR (Alt. 3 vs Alt. 2)

ΔCF(t=0): -55,000-(-35,000) = -20,000

ΔCF(t=1-4): 22,000-15,000 = 7,000/year

ΔCF(t=5): (22,000+5,000)-(15,000+3,000) = 9,000

$$PW_\Delta = -20{,}000+7{,}000(P/A,i,4)+9{,}000(P/F,i,5) = 0$$

Try i=25%: PW=-20,000+7,000(2.3616)+9,000(0.3277)=-20,000+16,531+2,949=-520

Try i=23%: PW=-20,000+7,000(2.4483)+9,000(0.3552)=-20,000+17,138+3,197=+335

$$\Delta IRR(3-2) = 23+2\times335/(335+520) = 23+0.79 = 23.79\% > MARR\ (15\%).\ Accept\ Alt.\ 3.$$

Final Answer: Select Alternative 3 (largest investment, highest total NPW).

Q15. Using incremental IRR, select the best project. MARR = 10%. [2076 Baisakh Q4a — 6 Marks]

Answer:

EOY	Project A	Project B	Project C
0	-15,000	-25,000	-40,000
1-5	5,500	9,000	13,500
Salvage (yr5)	1,000	1,500	2,000

MARR = 10%, n = 5 years. Uniform annual benefits + salvage at year 5.

$$(P/A,10\%,5)=3.7908,\quad (P/F,10\%,5)=0.6209$$

Step 1: Check each alternative against MARR

$$PW_A@10\% = -15{,}000+5{,}500(3.7908)+1{,}000(0.6209) = -15{,}000+20{,}849+621 = +6{,}470 > 0\ \checkmark$$

$$PW_B@10\% = -25{,}000+9{,}000(3.7908)+1{,}500(0.6209) = -25{,}000+34{,}117+931 = +10{,}048 > 0\ \checkmark$$

$$PW_C@10\% = -40{,}000+13{,}500(3.7908)+2{,}000(0.6209) = -40{,}000+51{,}175+1{,}242 = +12{,}417 > 0\ \checkmark$$

All alternatives meet MARR. Order by investment: A < B < C.

Step 2: ΔIRR (B vs. A)

$$\Delta CF(t=0)=-10{,}000,\ \Delta CF(t=1-4)=3{,}500/yr,\ \Delta CF(t=5)=3{,}500+500=4{,}000$$

Try i=28%: PW=-10,000+3,500(2.7594)+4,000(0.3916)=-10,000+9,658+1,567=+1,225

Try i=35%: PW=-10,000+3,500(2.3852)+4,000(0.2230)=-10,000+8,348+892=-760

$$\Delta IRR(B-A) = 28+7\times1{,}225/(1{,}225+760) = 28+4.3 = 32.3\% > 10\%.\ Accept\ B.$$

Step 3: ΔIRR (C vs. B)

$$\Delta CF(t=0)=-15{,}000,\ \Delta CF(t=1-4)=4{,}500/yr,\ \Delta CF(t=5)=4{,}500+500=5{,}000$$

Try i=24%: PW=-15,000+4,500(2.7454)+5,000(0.3411)=-15,000+12,354+1,706=-940

Try i=21%: PW=-15,000+4,500(2.9260)+5,000(0.3855)=-15,000+13,167+1,928=+95

$$\Delta IRR(C-B) = 21+3\times95/(95+940) = 21+0.28 = 21.3\% > 10\%.\ Accept\ C.$$

Select Project C (largest investment). PW_C = Rs. 12,417 is also the highest NPW, confirming the decision.

SECTION E: Incremental B/C Ratio Analysis

Q16. Explain the procedure for incremental B/C analysis for mutually exclusive public sector alternatives. [2078 Chaitra Q4a — 4 Marks]

Answer:

Why Incremental B/C is Needed

For public sector mutually exclusive alternatives, it is incorrect to simply choose the alternative with the highest B/C ratio. A high B/C ratio on a small project may not create as much public benefit as a lower B/C ratio on a larger project. Incremental B/C analysis determines whether the additional cost of a more expensive alternative generates sufficient additional benefit.

Procedure for Incremental B/C Analysis:

1. Order Alternatives by Cost: Arrange in ascending order of total equivalent cost (PW or AW of all costs including initial investment, O&M, and disbenefits).
2. Calculate B/C Ratio for the Cheapest Alternative: If B/C ≥ 1.0, this alternative is justified and becomes the Defender. If B/C < 1.0, eliminate it and move to the next.
3. Compute Incremental Benefits and Costs (Challenger vs. Defender):

```
ΔB/C = (B_Challenger - B_Defender) / (C_Challenger - C_Defender)
                = Incremental Benefits / Incremental Costs
```

4. Apply Decision Rule:
   - If ΔB/C ≥ 1.0: The extra cost of the Challenger is justified by the extra benefit. Challenger becomes the new Defender.
   - If ΔB/C < 1.0: Extra cost is NOT justified. Defender remains the current best.
5. Repeat: Continue comparing successive challengers against the current defender until all alternatives are evaluated.
6. Select the Final Defender as the best alternative.

Note on Modified B/C: If using the modified B/C ratio format, incremental analysis uses:

```
Δ(Modified B/C) = (ΔB - Δ(O&M)) / ΔInvestment ≥ 1.0 to accept Challenger
```

Q17. Use incremental B/C ratio analysis to select the best public project. MARR = 8%. [2081 Chaitra Q4b — 6 Marks]

Answer:

	Project 1	Project 2	Project 3	Project 4
PW of Costs (Rs. Cr)	10	15	22	30
PW of Benefits (Rs. Cr)	12	16	25	33

Using PW-based B/C analysis. Order by cost: Project 1 < 2 < 3 < 4.

Step 1: B/C of cheapest alternative (Project 1)

```
B/C_1 = 12/10 = 1.20 ≥ 1.0  →  Project 1 is justified. Defender = Project 1.
```

Step 2: Incremental B/C (Project 2 vs Project 1)

```
ΔB/C = (16-12)/(15-10) = 4/5 = 0.80 < 1.0
```

Extra cost of Project 2 over Project 1 is NOT justified. Reject Project 2. Defender remains Project 1.

Step 3: Incremental B/C (Project 3 vs Project 1)

$$\Delta B/C = (25-12)/(22-10) = 13/12 = 1.083 \geq 1.0$$

Extra investment in Project 3 over Project 1 is justified. Accept Project 3. Defender = Project 3.

Step 4: Incremental B/C (Project 4 vs Project 3)

$$\Delta B/C = (33-25)/(30-22) = 8/8 = 1.00 \geq 1.0$$

ΔB/C = 1.00 exactly meets the threshold. Accept Project 4. Defender = Project 4.

Select Project 4 (Final Defender). The incremental benefits of Project 4 over Project 3 exactly justify the incremental cost.

Q18. Select the best alternative using incremental conventional B/C ratio at MARR = 10%. [2080 Ashwin Q5a, 2079 Jestha Q5a — 8 Marks]

Answer:

	Alt. A	Alt. B	Alt. C	Alt. D
Initial Cost (Rs.)	30,000	50,000	75,000	1,00,000
Annual Benefit (Rs.)	8,000	13,500	19,000	24,000
Annual O&M (Rs.)	1,500	2,500	3,500	4,500
Salvage (Rs.)	3,000	5,000	7,000	9,000
Life (years)	10	10	10	10

MARR = 10%, n = 10 years.

$$(P/A,10\%,10)=6.1446, \quad (P/F,10\%,10)=0.3855$$

Step 1: Compute PW of Costs and Benefits

PW of Costs = Initial Investment + PW(Annual O&M) - PW(Salvage)

PW of Benefits = PW(Annual Benefit)

	Alt. A	Alt. B	Alt. C	Alt. D
PW of Annual Benefit	49,157	82,952	116,747	147,470
PW of Annual O&M	9,217	15,362	21,506	27,651
PW of Salvage	1,157	1,928	2,699	3,470
PW of Costs = Inv+OM-Salv	38,060	63,434	95,807	1,24,181
Conventional B/C	1.292	1.308	1.219	1.188

Calculation detail for Alt. A: B = 8,000×6.1446 = 49,157; O&M = 1,500×6.1446 = 9,217; S = 3,000×0.3855 = 1,157; Cost = 30,000+9,217-1,157 = 38,060; B/C = 49,157/38,060 = 1.292

All B/C > 1.0. All alternatives justified individually. Order by cost: A < B < C < D.

Step 2: ΔIRR(B vs A)

$$\Delta B = 82{,}952-49{,}157 = 33{,}795; \quad \Delta C = 63{,}434-38{,}060 = 25{,}374$$

$$\Delta B/C = 33{,}795/25{,}374 = 1.332 \geq 1.0 \rightarrow \text{Accept B. Defender = B.}$$

Step 3: ΔB/C(C vs B)

$$\Delta B = 116{,}747-82{,}952 = 33{,}795; \quad \Delta C = 95{,}807-63{,}434 = 32{,}373$$

$$\Delta B/C = 33{,}795/32{,}373 = 1.044 \geq 1.0 \rightarrow \text{Accept C. Defender = C.}$$

Step 4: ΔB/C(D vs C)

```
ΔB = 147,470-116,747 = 30,723;  ΔC = 1,24,181-95,807 = 28,374
ΔB/C = 30,723/28,374 = 1.083 ≥ 1.0  →  Accept D. Defender = D.
```

Select Alternative D. Each increment of investment generates more than Re. 1 of benefit per Re. 1 of cost, justifying the full investment in D.

SECTION F: Capitalized Cost Comparison

Q19. Define capitalized cost. What is the formula for capitalized cost? Compare two permanent alternatives using capitalized cost at MARR = 8%. [2079 Ashwin Q5b, 2078 Baishakh Q4a — 5 Marks]

Answer:

Capitalized Cost — Definition

Capitalized Cost (CC) is the Present Worth of an alternative that is assumed to last forever (or for a very long time — typically for infrastructure, bridges, dams, tunnels). It represents the sum of money that, invested today at MARR, would generate sufficient interest income to fund all costs of the project indefinitely without depleting the principal.

```
For a permanent annual cost A:  CC = A / i
For an initial investment P with perpetual annual costs A:  CC = P + A/i
For repeating costs C every k years:  CC = C / [(1+i)^k - 1]
```

This formula comes from the limit of (P/A, i, n) as n → ∞: (P/A, i, ∞) = 1/i

Numerical Comparison:

	Structure P	Structure Q
Initial Cost (Rs.)	5,00,000	8,00,000
Annual Maintenance (Rs.)	25,000	15,000
Major Overhaul every k years (Rs. / period)	50,000 every 5 years	80,000 every 10 years
Life	Permanent	Permanent

MARR = 8%

Capitalized Cost of Structure P:

```
CC_P = Initial Cost + CC(Annual Maintenance) + CC(Overhaul)
CC(Annual Maintenance) = 25,000 / 0.08 = Rs. 3,12,500
CC(Overhaul every 5 yrs) = 50,000 / [(1.08)^5 - 1] = 50,000 / 0.4693 = Rs. 1,06,541
CC_P = 5,00,000 + 3,12,500 + 1,06,541 = Rs. 9,19,041
```

Capitalized Cost of Structure Q:

```
CC(Annual Maintenance) = 15,000 / 0.08 = Rs. 1,87,500
CC(Overhaul every 10 yrs) = 80,000 / [(1.08)^10 - 1] = 80,000 / 1.1589 = Rs. 69,034
```

CC_Q = 8,00,000 + 1,87,500 + 69,034 = Rs. 10,56,534

CC_P (Rs. 9,19,041) < CC_Q (Rs. 10,56,534). Select Structure P as it has lower total capitalized cost.

Q20. Two alternatives are being compared for a permanent dam structure. Find capitalized costs and recommend the better alternative at MARR = 10%. [2077 Chaitra Q5a — 5 Marks]

Answer:

	Design A	Design B
Initial Construction Cost (Rs.)	40,00,000	65,00,000
Annual Operating Cost (Rs.)	2,00,000	1,20,000
Periodic Renovation (Rs. / cycle)	8,00,000 every 20 years	6,00,000 every 25 years
Life	Permanent	Permanent

MARR = 10%

Capitalized Cost of Design A:

CC(Annual O&M) = 2,00,000 / 0.10 = Rs. 20,00,000

CC(Renovation every 20 yrs) = 8,00,000 / [(1.10)^20 - 1] = 8,00,000 / (6.7275-1) = 8,00,000 / 5.7275

= Rs. 1,39,654

CC_A = 40,00,000 + 20,00,000 + 1,39,654 = Rs. 61,39,654

Capitalized Cost of Design B:

CC(Annual O&M) = 1,20,000 / 0.10 = Rs. 12,00,000

CC(Renovation every 25 yrs) = 6,00,000 / [(1.10)^25 - 1] = 6,00,000 / (10.8347-1) = 6,00,000 / 9.8347

= Rs. 61,008

CC_B = 65,00,000 + 12,00,000 + 61,008 = Rs. 77,61,008

CC_A (Rs. 61,39,654) < CC_B (Rs. 77,61,008). Select Design A.

Q21. Find capitalized cost of a project with initial investment Rs. 5,00,000, annual operation cost Rs. 40,000, and overhauling cost Rs. 1,00,000 every 5 years. Interest rate = 12% per year. [2075 Bhadra Q5a, 2072 Magh Q5a — 5 Marks]

Answer:

Given: P = Rs. 5,00,000, Annual O&M = Rs. 40,000, Overhaul = Rs. 1,00,000 every 5 years, i = 12%

Capitalized Cost of Annual O&M:

CC(O&M) = A / i = 40,000 / 0.12 = Rs. 3,33,333

Capitalized Cost of Recurring Overhaul (every 5 years):

CC(Overhaul) = C / [(1+i)^k - 1] = 1,00,000 / [(1.12)^5 - 1]

$$(1.12)^5 = 1.7623$$

$$CC(Overhaul) = 1,00,000 / (1.7623 - 1) = 1,00,000 / 0.7623 = Rs.\ 1,31,186$$

Total Capitalized Cost:

$$CC_total = P + CC(O\&M) + CC(Overhaul)$$

$$= 5,00,000 + 3,33,333 + 1,31,186 = Rs.\ 9,64,519$$

Capitalized Cost = Rs. 9,64,519.

Interpretation: Rs. 9,64,519 invested today at 12% would generate sufficient interest to fund the initial construction, annual O&M, and all future 5-yearly overhauls indefinitely.

SECTION G: Comprehensive Alternative Comparison Problems

Q22. Four mutually exclusive alternatives are available. Using PW method, find the best alternative at MARR = 12%. [2081 Ashwin Q5b — 6 Marks]

Answer:

EOY	Alt. A	Alt. B	Alt. C	Alt. D
0	-50,000	-80,000	-1,10,000	-1,40,000
1-8	14,000	22,000	30,000	37,000
Salvage (yr8)	5,000	8,000	10,000	12,000

MARR = 12%, n = 8 years (equal lives — direct PW comparison)

$$(P/A,12\%,8)=4.9676, \quad (P/F,12\%,8)=0.4039$$

$$PW_A = -50,000+14,000(4.9676)+5,000(0.4039) = -50,000+69,546+2,020 = +21,566$$

$$PW_B = -80,000+22,000(4.9676)+8,000(0.4039) = -80,000+1,09,287+3,231 = +32,518$$

$$PW_C = -1,10,000+30,000(4.9676)+10,000(0.4039) = -1,10,000+1,49,028+4,039 = +43,067$$

$$PW_D = -1,40,000+37,000(4.9676)+12,000(0.4039) = -1,40,000+1,83,801+4,847 = +48,648$$

PW_D (Rs. 48,648) is the highest. All PW > 0. Select Alternative D.

Q23. Compare the following alternatives using AW method at MARR = 10%. Lives are unequal. [2079 Chaitra Q5a — 8 Marks]

Answer:

	Alt. X	Alt. Y	Alt. Z
Initial Cost (Rs.)	1,20,000	1,80,000	2,50,000
Annual Revenue (Rs.)	40,000	55,000	72,000
Annual O&M (Rs.)	10,000	13,000	16,000
Salvage Value (Rs.)	15,000	25,000	35,000

Life (years)	4	6	8

MARR = 10%. Unequal lives. Use AW method (repeatability assumption).

```
(A/P,10%,4)=0.31547   (A/F,10%,4)=0.21547
(A/P,10%,6)=0.22961   (A/F,10%,6)=0.12961
(A/P,10%,8)=0.18744   (A/F,10%,8)=0.08744
```

AW of Alt. X (4 years):

```
AW_X = -1,20,000(0.31547)+15,000(0.21547)+(40,000-10,000)
       = -37,856+3,232+30,000 = -Rs. 4,624
```

AW of Alt. Y (6 years):

```
AW_Y = -1,80,000(0.22961)+25,000(0.12961)+(55,000-13,000)
       = -41,330+3,240+42,000 = +Rs. 3,910
```

AW of Alt. Z (8 years):

```
AW_Z = -2,50,000(0.18744)+35,000(0.08744)+(72,000-16,000)
       = -46,860+3,060+56,000 = +Rs. 12,200
```

AW_X < 0 (does not meet MARR, REJECT). AW_Y = Rs. 3,910 and AW_Z = Rs. 12,200 > 0. AW_Z is highest. Select Alternative Z.

Q24. Select the best alternative from the following, using AW and PW methods. MARR = 10%. [2070 Magh Q5a, 2074 Bhadra Q5a — 8 Marks]

Answer:

	Option I	Option II	Option III
First Cost (Rs.)	1,00,000	1,50,000	2,00,000
Annual Benefit (Rs.)	30,000	40,000	55,000
Annual O&M (Rs.)	5,000	7,000	9,000
Salvage (Rs.)	10,000	15,000	20,000
Life (years)	5	5	5

MARR = 10%, n = 5 years (equal lives)

```
(P/A,10%,5)=3.7908,   (P/F,10%,5)=0.6209
(A/P,10%,5)=0.26380,  (A/F,10%,5)=0.16380
```

PW Method:

```
PW_I   = -1,00,000+(30,000-5,000)(3.7908)+10,000(0.6209) = -
         1,00,000+94,770+6,209 = +Rs. 979
PW_II  = -1,50,000+(40,000-7,000)(3.7908)+15,000(0.6209) = -
         1,50,000+1,25,096+9,314 = -Rs. 15,590
PW_III = -2,00,000+(55,000-9,000)(3.7908)+20,000(0.6209) = -
         2,00,000+1,74,377+12,418 = -Rs. 13,205
```

Only Option I has positive PW. Options II and III do not meet MARR.

AW Method (confirming):

```
AW_I   = -1,00,000(0.26380)+10,000(0.16380)+(30,000-5,000) = -
                26,380+1,638+25,000 = +Rs. 258
AW_II  = -1,50,000(0.26380)+15,000(0.16380)+(40,000-7,000) = -
               39,570+2,457+33,000 = -Rs. 4,113
AW_III = -2,00,000(0.26380)+20,000(0.16380)+(55,000-9,000) = -
               52,760+3,276+46,000 = -Rs. 3,484
```

Both PW and AW confirm: Only Option I meets MARR. Select Option I (AW = Rs. 258 > 0, PW = Rs. 979 > 0). Options II and III do not achieve the 10% minimum return.

Q25. From the three alternatives given, use AW method to select the best at MARR = 12%. [2075 Bhadra Q4a — 6 Marks]

Answer:

	Machine A	Machine B	Machine C
Capital Investment (Rs.)	2,50,000	3,50,000	5,00,000
Annual Revenue (Rs.)	75,000	1,00,000	1,25,000
Annual Expenses (Rs.)	20,000	25,000	30,000
Salvage Value (Rs.)	30,000	50,000	75,000
Life (years)	5	8	10

MARR = 12%. Unequal lives — use AW for one life cycle each.

```
(A/P,12%,5)=0.27741   (A/F,12%,5)=0.15741
(A/P,12%,8)=0.20130   (A/F,12%,8)=0.08130
(A/P,12%,10)=0.17698  (A/F,12%,10)=0.05698
```

AW of Machine A (5 years):

```
AW_A = -2,50,000(0.27741)+30,000(0.15741)+(75,000-20,000)
        = -69,353+4,722+55,000 = -Rs. 9,631
```

AW of Machine B (8 years):

```
AW_B = -3,50,000(0.20130)+50,000(0.08130)+(1,00,000-25,000)
        = -70,455+4,065+75,000 = +Rs. 8,610
```

AW of Machine C (10 years):

```
AW_C = -5,00,000(0.17698)+75,000(0.05698)+(1,25,000-30,000)
        = -88,490+4,274+95,000 = +Rs. 10,784
```

AW_A < 0 (reject). AW_C (Rs. 10,784) > AW_B (Rs. 8,610) > 0. Select Machine C.

SECTION H: Public Project Selection — Incremental B/C with Full Cash Flows

Q26. Three public-sector road projects are being evaluated. Use incremental B/C ratio to identify the best project. MARR = 8%, n = 20 years. [2080 Chaitra Q5a — 8 Marks]

Answer:

	Road A	Road B	Road C
Construction Cost (Rs. Cr)	20	35	55
Annual Maintenance (Rs. Cr)	1.0	1.5	2.0
Annual User Benefits (Rs. Cr)	4.5	7.5	11.0
Salvage Value (Rs. Cr)	2	4	6

MARR = 8%, n = 20 years.

```
(P/A,8%,20)=9.8181,   (P/F,8%,20)=0.2145
```

Step 1: Calculate PW of Costs and Benefits for each road

PW of Total Costs = Construction + PW(Maintenance) - PW(Salvage)

Road A: Costs = 20 + 1.0(9.8181) - 2(0.2145) = 20 + 9.818 - 0.429 = Rs. 29.389 Cr

Road B: Costs = 35 + 1.5(9.8181) - 4(0.2145) = 35 + 14.727 - 0.858 = Rs. 48.869 Cr

Road C: Costs = 55 + 2.0(9.8181) - 6(0.2145) = 55 + 19.636 - 1.287 = Rs. 73.349 Cr

PW of Benefits = Annual user benefits × (P/A,8%,20)

Road A: Benefits = 4.5 × 9.8181 = Rs. 44.181 Cr

Road B: Benefits = 7.5 × 9.8181 = Rs. 73.636 Cr

Road C: Benefits = 11.0 × 9.8181 = Rs. 107.999 Cr

Step 2: Check individual B/C ratios

```
B/C_A = 44.181/29.389 = 1.503 ≥ 1.0  ✓  Defender = A
B/C_B = 73.636/48.869 = 1.507 ≥ 1.0  ✓  (also viable but compare incrementally)
B/C_C = 107.999/73.349 = 1.473 ≥ 1.0  ✓
```

Step 3: Incremental B/C (B vs. A)

```
ΔB/C = (73.636-44.181)/(48.869-29.389) = 29.455/19.480 = 1.512 ≥ 1.0  → Accept B
```

Defender = B

Step 4: Incremental B/C (C vs. B)

```
ΔB/C = (107.999-73.636)/(73.349-48.869) = 34.363/24.480 = 1.404 ≥ 1.0  → Accept C
```

Select Road C (Final Defender). The incremental benefits of each successive road justify the incremental cost. Road C provides the maximum net public benefit.

QUICK REFERENCE: When to Use Each Comparison Method

Situation	Best Method	Key Rule

Equal lives, revenue alternatives	PW or AW	Select highest positive PW or AW
Equal lives, cost-only alternatives	AW or PW	Select least-cost (most negative AW is worst)
Unequal lives, revenue	AW (one cycle) or PW (LCM)	AW over one cycle = AW over LCM
Unequal lives, cost-only	AW over one cycle	Select lowest annual cost
Fixed study period	PW over study period	Estimate market value for truncated alts
Mutually exclusive with IRR	Incremental IRR (ΔIRR)	Accept if ΔIRR ≥ MARR
Public sector alternatives	Incremental B/C	Accept if ΔB/C ≥ 1.0
Very long or permanent life	Capitalized Cost	CC = P + A/i + C/[(1+i)^k-1]
Independent alternatives	Individual IRR or PW	Accept each if IRR ≥ MARR or PW ≥ 0

Important Reminders:

- Always include the Do-Nothing alternative in any comparison set.
- For AW method with unequal lives: AW of one cycle is sufficient (no LCM needed).
- For PW method with unequal lives: Use LCM or co-terminated study period.
- For incremental analysis: Order alternatives by ascending initial cost; start with smallest investment.
- An alternative with NPW < 0 or B/C < 1.0 individually should be eliminated before incremental analysis.
- Incremental IRR analysis and NPW method will ALWAYS give the same selection decision.

—— End of Chapter 4 Questions & Answers ——